

NIB Social Hub – Complete Beginner & Master Guide (Localhost & PostgreSQL)

This guide is designed for beginners and also includes recommended advanced/master practices for running NIB Social Hub effectively.

1 Install PostgreSQL on Windows

1. Go to <https://www.postgresql.org/download/windows/>
2. Download the **EDB PostgreSQL Installer**
3. Run the installer, select:
4. PostgreSQL Server
5. pgAdmin 4
6. Command Line Tools (psql)
7. Set a strong password for postgres
8. Keep default port 5432

Tip: If psql is not recognized, add to PATH:

```
C:\Program Files\PostgreSQL\16\bin
```

2 Test PostgreSQL

Open Command Prompt:

```
psql -U postgres
```

Enter password (typing is invisible). You should see:

```
postgres=#
```

3 Create Database

```
CREATE DATABASE nib_social_hub;  
\c nib_social_hub
```

Now you are inside the database:

```
nib_social_hub=#
```

4 Create Tables

Users Table

```
CREATE TABLE users (  
    id SERIAL PRIMARY KEY,  
    full_name VARCHAR(150) NOT NULL,  
    email VARCHAR(150) UNIQUE NOT NULL,  
    phone VARCHAR(20),  
    branch VARCHAR(100),  
    role VARCHAR(50),  
    password VARCHAR(255),  
    created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP  
);
```

Posts Table

```
CREATE TABLE posts (  
    id SERIAL PRIMARY KEY,  
    user_id INT REFERENCES users(id) ON DELETE CASCADE,  
    content TEXT NOT NULL,  
    image_url TEXT,  
    created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP  
);
```

Comments Table

```
CREATE TABLE comments (  
    id SERIAL PRIMARY KEY,  
    post_id INT REFERENCES posts(id) ON DELETE CASCADE,  
    user_id INT REFERENCES users(id) ON DELETE CASCADE,  
    comment TEXT NOT NULL,  
    created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP  
);
```

Likes Table

```
CREATE TABLE likes (  
    id SERIAL PRIMARY KEY,  
    post_id INT REFERENCES posts(id) ON DELETE CASCADE,
```

```
    user_id INT REFERENCES users(id) ON DELETE CASCADE
);
```

Check tables:

```
\dt
```

5 Insert a Test User

```
INSERT INTO users (full_name, email, phone, branch, role, password)
VALUES ('Abiy Hailemichael', 'abiy@nibbank.com', '0912345678', 'Addis Ababa
Main', 'Admin', '1234');
```

Check data:

```
SELECT * FROM users;
```

6 Connect VS Code Project to PostgreSQL

Install Node.js PostgreSQL driver:

```
npm install pg bcrypt
```

Create `db.js`:

```
const { Pool } = require('pg');
const pool = new Pool({
  user: 'postgres',
  host: 'localhost',
  database: 'nib_social_hub',
  password: 'YourPasswordHere',
  port: 5432,
});
module.exports = pool;
```

7 Authentication (Signup/Login)

Use bcrypt to hash passwords:

```
const bcrypt = require('bcrypt');
const hashedPassword = await bcrypt.hash(userPassword, 10);
const match = await bcrypt.compare(inputPassword, storedHashedPassword);
```

8 Basic CRUD Operations

Create a Post

```
await pool.query(
  'INSERT INTO posts (user_id, content, image_url) VALUES ($1, $2, $3)',
  [userId, content, imageUrl]
);
```

Fetch Posts

```
const posts = await pool.query('SELECT * FROM posts ORDER BY created_at DESC');
```

Comment on a Post

```
await pool.query(
  'INSERT INTO comments (post_id, user_id, comment) VALUES ($1, $2, $3)',
  [postId, userId, commentText]
);
```

Like a Post

```
await pool.query(
  'INSERT INTO likes (post_id, user_id) VALUES ($1, $2)',
  [postId, userId]
);
```

9 Recommended Master / Advanced Features

- **Role-Based Access Control (RBAC):** Admin, Branch Officer, Employee, Guest
- **Activity Logs:** Track user activity in a separate `logs` table
- **Notifications:** Push notifications or alerts for posts/comments
- **Search & Filter:** Full-text search on posts, filter by branch or role
- **File Uploads:** Store images/files in cloud (S3, Firebase) with database reference
- **Pagination & Infinite Scroll:** For posts/comments
- **API Layer:** Use REST or GraphQL for front-end/backend separation
- **Environment Variables:** Store DB credentials, JWT secret, and keys in `.env`

- **JWT Authentication:** Secure token-based login for frontend
 - **Backup & Restore:** Regular dump of PostgreSQL database
 - **Security:** SSL connection, input validation, password hashing, rate limiting
 - **Deployment:** Use Docker or cloud hosting (AWS, Heroku, Railway) for production
 - **CI/CD Pipeline:** Automate deployment and testing for the project
-

10 Deployment Notes

- Local development: `localhost:3000`
 - Production:
 - Cloud DB (AWS RDS, Heroku Postgres)
 - Environment variables for credentials
 - Hash passwords
 - Set proper CORS
 - SSL connections
-

1 1 Best Practices

- Never store plain passwords
 - Validate user inputs
 - Role-based access (Admin, Employee, Branch Officer)
 - Regular database backup
 - Prevent SQL injection
 - Use secure password policies and password hashing
 - Keep dependencies updated and patched
 - Maintain clear code structure and modularization
-

✓ This guide now covers beginner steps and recommended advanced/master features for NIB Social Hub. It can be used as a **PDF or printed manual** for learning and deployment purposes.